Signal Processing in Practice

Assignment 1: Image Denoising
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1 Introduction

Image denoising is a fundamental problem in signal processing. The goal is to recover a clean signal $x(m,n)$ from a noisy observation $y(m,n) = x(m,n) + \eta(m,n)$, where η represents noise. This assignment explores two approaches:

1. **Experiment 1:** Low Pass Gaussian Filter.
2. **Experiment 2:** Bilateral Filter.

2 Experiment 1: Optimal Gaussian Filtering

2.1 Methodology

The Gaussian low-pass filter is a linear convolution operator. The denoised image $\hat{x}(m,n)$ is computed as:

$$\hat{x}(m,n) = \sum_{k=-P}^{P} \sum_{l=-P}^{P} w(k,l) y(m+k, n+l) \tag{1}$$

The weights $w(k,l)$ are derived from a 2D Gaussian distribution:

$$w(k,l) = C \cdot \exp\left(-\frac{k^2 + l^2}{2\sigma^2}\right) \tag{2}$$

where P defines the window radius, σ is the standard deviation, and C is a normalization constant ensuring $\sum\sum w(k,l) = 1$.

We utilized a window size of 11×11 ($P = 5$) and optimized σ from the set $\{0.1, 1, 2, 4, 8\}$ to minimize the Mean Squared Error (MSE) with respect to the reference image `img167.bmp`.

2.2 Results

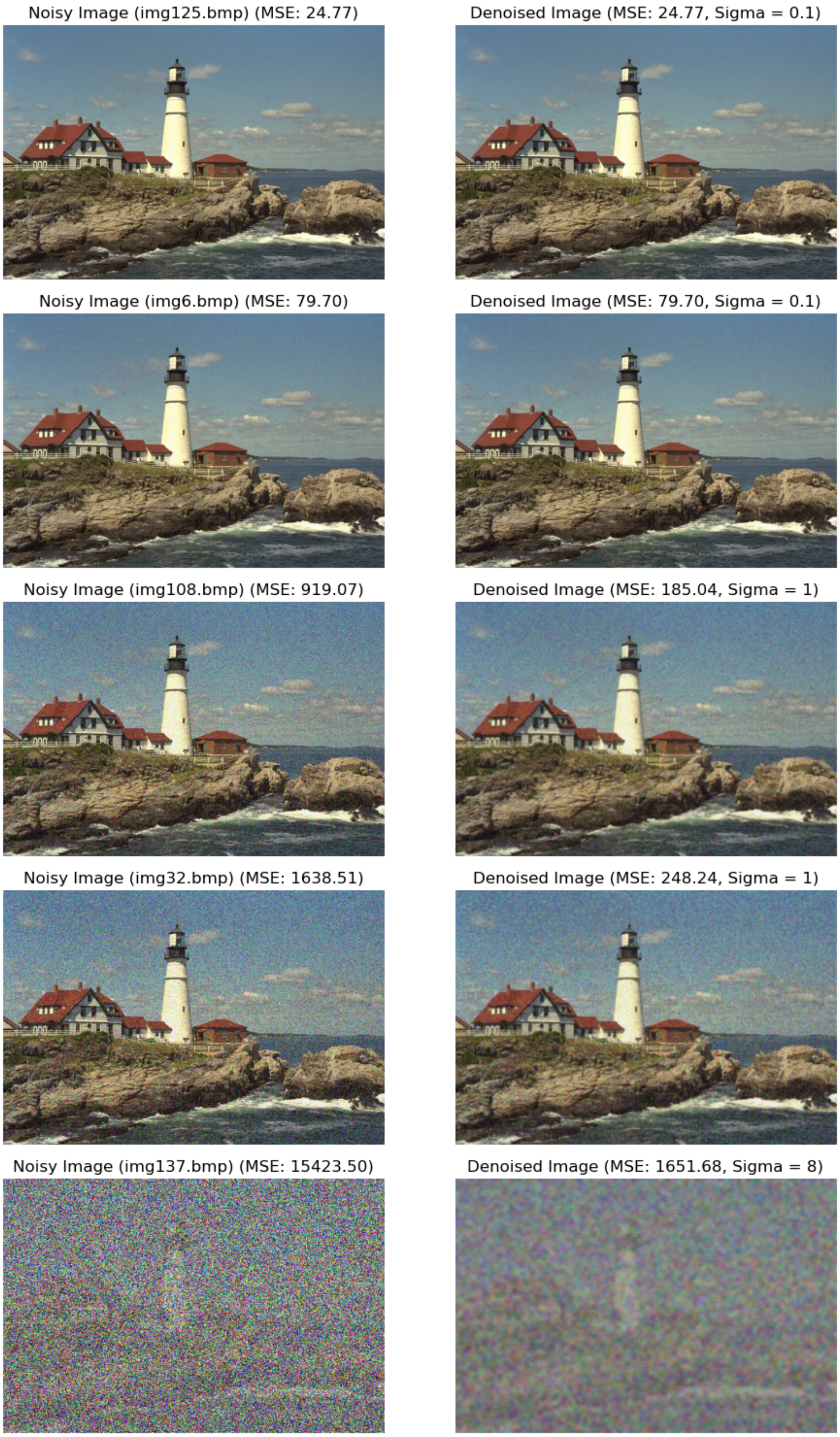

Figure 1: The figure displays the noisy input images alongside their optimally denoised counterparts. The optimal σ increases with the noise level.

The quantitative results are summarized in Table 1.

Table 1: Optimal Gaussian σ and Minimum MSE for varying noise levels.

Image Index	Noise Level	Optimal σ	Min MSE
img125	Low	0.1	**24.77**
img6	Low	0.1	**79.70**
img108	Medium	1.0	**185.04**
img32	Medium	1.0	**248.24**
img137	Severe	8.0	**1651.68**

2.3 Analysis

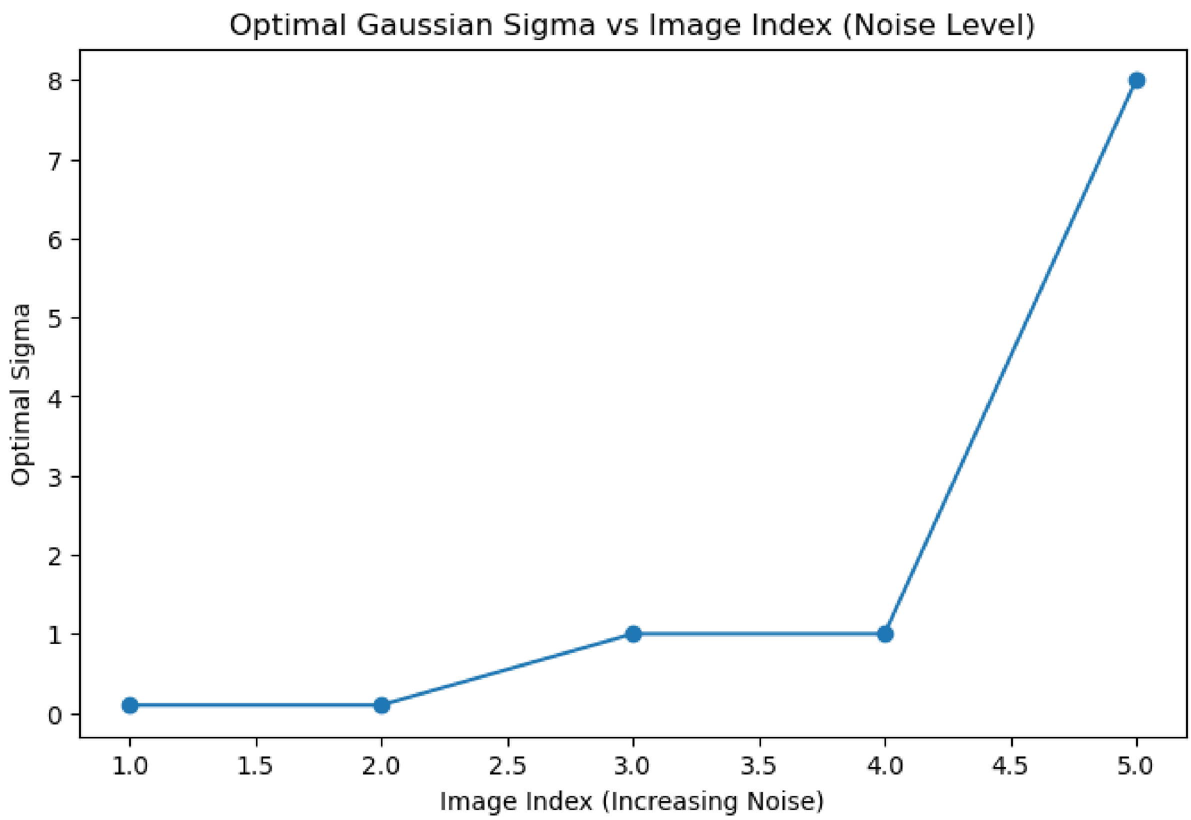

Figure 2: Optimal Standard Deviation (σ) vs. Noisy Image Index.

Observation: The curve in Fig: 2 is monotonically increasing.
Reasoning: Gaussian filtering reduces noise variance by averaging N pixels, reducing the noise standard deviation by a factor of $\sqrt{N}$.

- For low noise, a small σ is sufficient. Using a large σ would introduce excessive blurring without significantly improving the noise reduction.

- For high noise, the noise variance dominates. To minimize total MSE, a larger σ is required to average over a wider neighborhood, accepting some blur to effectively suppress the heavy grain.

3 Experiment 2: Bilateral Filtering

3.1 Methodology

The Bilateral filter is a non-linear, edge-preserving smoothing technique. Unlike the standard Gaussian filter, which relies solely on spatial proximity, the Bilateral filter weights neighbors based on both their geometric closeness and their photometric (intensity) similarity.

For a noisy image $y(m,n)$, the filtered output $\hat{x}(m,n)$ is defined as:

$$\hat{x}(m,n) = \frac{1}{C(m,n)} \sum_{k=-P}^{P} \sum_{l=-P}^{P} G(k,l) \cdot H(y(m,n) - y(m+k,n+l)) \cdot y(m+k,n+l) \quad (3)$$

where:

- P represents the window radius (where window size is $(2P+1) \times (2P+1)$).
- $y(m,n)$ is the intensity of the center pixel, and $y(m+k,n+l)$ is the intensity of the neighbor at offset (k,l).

The weighting kernels are defined as follows:

1. Spatial Domain Kernel (G): This term assigns higher weights to pixels that are spatially close to the center (m,n). It is a standard Gaussian function of the spatial distance:

$$G(k,l) = \exp\left(-\frac{k^2+l^2}{2\sigma_G^2}\right) \quad (4)$$

where σ_G controls the spatial extent of the filter.

2. Range (Intensity) Domain Kernel (H): This term assigns higher weights to pixels that have a similar intensity value to the center pixel. It acts as an edge-stopping function:

$$H(\delta) = \exp\left(-\frac{\delta^2}{2\sigma_H^2}\right) \quad (5)$$

where $\delta = y(m,n) - y(m+k,n+l)$ is the intensity difference, and σ_H controls the sensitivity to edges.

3. Normalization Constant (C): To preserve the local mean brightness, the weights are normalized by $C(m,n)$:

$$C(m,n) = \sum_{k=-P}^{P} \sum_{l=-P}^{P} G(k,l) \cdot H(y(m,n) - y(m+k,n+l)) \quad (6)$$

3.1.1 Parameter Selection

The filter behavior is governed by two parameters:

- σ_G (Spatial Sigma): Determines the neighborhood size for smoothing. A larger σ_G combines values from a wider area.
- σ_H (Range Sigma): Determines edge sensitivity. If the intensity difference δ is large (e.g., at an edge), $H(\delta) \approx 0$, effectively preserving the sharp transition.

3.2 Visual Comparison

We applied the Bilateral filter to `noisybook.png` using parameters $\sigma_s = 4.0$ and $\sigma_r = 30.0$.

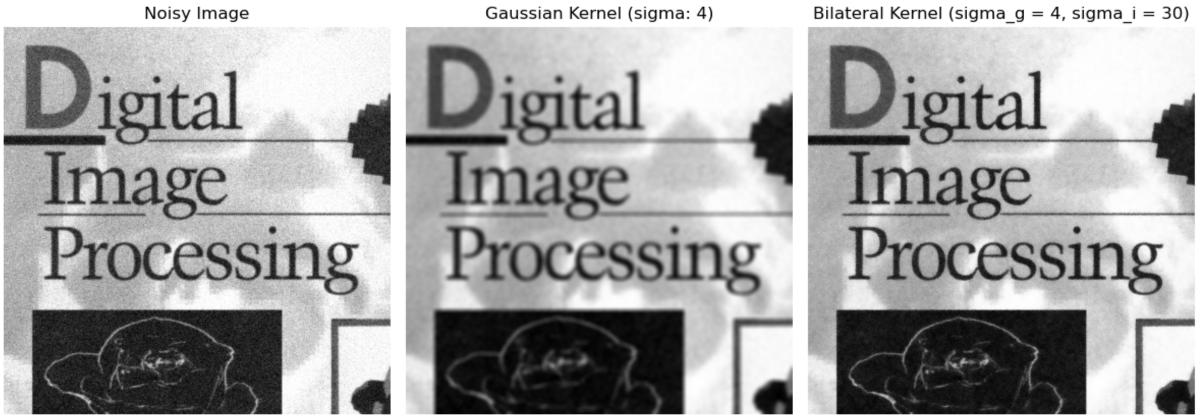

Figure 3: Comparison on `noisybook.png`. (Left) Noisy Input. (Center) Gaussian Filter ($\sigma = 4$). (Right) Bilateral Filter.

3.3 Qualitative Analysis

- **Gaussian Filter:** The linear filter treats edges (text-to-background transitions) as noise. By averaging black text pixels with the white background, it produces a "smudged" result, reducing readability.
- **Bilateral Filter:** The range term H_{σ_r} acts as an edge-stop function. When the filter window crosses a text boundary, the intensity difference is large, causing the weight to drop to zero. This prevents averaging across the edge, resulting in crisp text and a smooth background.